Keegan Sanctis

College Park, MD, USA | +1 (227) 248-1667 | keegansanctis@gmail.com | LinkedIn | Web Portfolio | GitHub

EDUCATION

Master of Science in Applied Machine Learning — **Anticipated Graduation: May 2027**
University of Maryland, College Park, MD, USA — **Current GPA: 4.0/4.0**

Bachelor of Computer Science, Minor in Economics — **Graduation Year: 2023**
University of Waterloo, Waterloo, ON, Canada

TECHNICAL SKILLS

Software Engineering: React, Angular, Docker, Kubernetes, CI/CD, Automated Testing, Object-Oriented Programming

Programming: Python, JavaScript, C++, C, SQL

Backend & Databases: Node.js, Express.js, PostgreSQL

Machine Learning & Data Science: scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, Keras, Statistical Modeling

WORK EXPERIENCE

Software & Business Developer | RoyalTech Electrical & IntelliLedge — **Jul 2024 – Present**

- Designed and built IntelliLedge, a full-stack SaaS POS and accounting platform using React, Node.js, Express, PostgreSQL
- Implemented role-based and location-based access controls to restrict sensitive operations by user role and location
- Automated data reuse across quotations, delivery orders, and invoices, reducing manual data entry by ~65%
- Developing an ML-based engine to recommend quotation pricing using historical quotation acceptance data

Software Developer (Co-op) | D2L, Kitchener, ON, Canada — **Sep 2022 – Dec 2022**

- Built dropdown and filtering UI components for Brightspace, used by 1,300+ organizations and millions of learners
- Extended a shared UI component library used by 6 teams based on feedback from internal UX stakeholders
- Improved accessibility and usability features across Brightspace, reducing user-reported issues by 15%

Software Engineering Consulting (Co-op) | Capco, Toronto, ON, Canada — **Jan 2022 – Apr 2022**

- Migrated BMO Harris Bank's online banking app from Angular 5 to 13 and Ionic 3 to 5, reducing page load times by 30%
- Developed the login and landing dashboard pages for the banking app, resolving migration-related compatibility issues
- Resolved post-upgrade bugs with QA and maintained a suite of regression tests throughout the framework migration

Software Developer (Co-op) | Marsh McLennan, Toronto, ON, Canada — **May 2021 – Aug 2021**

- Rebuilt the corporate information systems portal with React and Ionic, increasing session duration by 58%
- Implemented monitoring and alerting for Docker/Kubernetes services to help detect system issues and reduce downtime
- Built a CodeceptJS and Playwright regression suite, reducing testing time by 42% and supporting continuous integration

PROJECTS

Safe Reinforcement Learning (RL) for Insulin Dosing | University of Maryland — **Fall 2026**

- Implementing offline RL algorithms (behavior cloning, TD3+BC, CQL) for constrained insulin dosing in Type 1 diabetes
- Building an evaluation pipeline for dosing policies across 30 simulated patients spanning adults, adolescents, and children
- Evaluating cross-cohort generalization against a target of over 70% time in range and under 4% hypoglycemia
- Formulating insulin dosing as a constrained Markov DP and comparing learned policies with PID and basal-bolus controllers

Portfolio Optimization for Inventory Management | University of Maryland — **Spring 2026**

- Adapted Markowitz mean-variance portfolio optimization to allocate inventory across multiple SKUs
- Formulated and solved a convex inventory optimization model in CVXPY across 3,856 SKUs from RoyalTech's dataset
- Built a data pipeline to process ~391,000 rows of historical sales and purchase records and estimate item-level demand
- Generated efficient frontiers across 30 budget levels, reducing modeled costs by 22.61% versus uniform allocation

Gun Ownership and Shooting Incident Analysis | University of Maryland — **Fall 2025**

- Analyzed 8,098 U.S. shooting incidents and gun ownership rates, trained decision tree classifiers with 91.6% test accuracy
- Compared state-level gun ownership trends from 1980-2016 against shooting incidents, examining their relationship over time